# Question ID 2e92cc21

| Assessment | Test | Domain | Skill | Difficulty |
|---|---|---|---|---|
| SAT | Math | Problem-Solving and Data Analysis | Percentages | |

**ID: 2e92cc21**

The number $a$ is $110\%$ greater than the number $b$. The number $b$ is $90\%$ less than $47$. What is the value of $a$?

**ID: 2e92cc21 Answer**

**Correct Answer:**
9.87, 987/100

**Rationale**

The correct answer is $9.87$. It's given that the number $a$ is $110\%$ greater than the number $b$. It follows that $a = \left(1 + \frac{110}{100}\right)b$, or $a = 2.1b$. It's also given that the number $b$ is $90\%$ less than $47$. It follows that $b = \left(1 - \frac{90}{100}\right)(47)$, or $b = 0.1(47)$, which yields $b = 4.7$. Substituting $4.7$ for $b$ in the equation $a = 2.1b$ yields $a = 2.1(4.7)$, which is equivalent to $a = 9.87$. Therefore, the value of $a$ is $9.87$.

**Question Difficulty:**
Hard